

Golf Launch Monitor - Milestone 3: Club/Ball Separation Logic

Complete MATLAB Implementation Prompt for Cursor AI

PROJECT CONTEXT

You are implementing Milestone 3 of a golf launch monitor DSP system that processes radar I/Q data to separate golf club head velocities from ball velocities. This milestone must be completely self-contained and independent from previous milestones, focusing specifically on the critical challenge of club/ball separation that determines system accuracy.

TECHNICAL SPECIFICATIONS

Dataset Format:

- Binary files: `LOG[YYYYMMDD][HHMMSS]_[SEQUENCE].bin` (4-channel I/Q data)
- JSON metadata: Configuration parameters, TrackMan reference speeds embedded in filenames
- Sampling rate: 22.7 kHz, FFT size: 1024, Data step: 64
- Velocity range: 56.5-178.9 mph (ball), 45.7-120.4 mph (club head)
- Speed coefficient: 0.138888888888889 (frequency to mph conversion)

Hardware: SMR333 radar (24.125 GHz), Target MCUs: Renesas Rx65 / STM32U

MILESTONE 3 OBJECTIVES

Primary Goals:

1. **Impact Detection:** Identify precise moment of club-ball contact using energy burst analysis
2. **Timing-based Separation:** Use temporal characteristics to distinguish club vs ball velocity signatures
3. **Classification Framework:** Implement robust decision algorithms for club/ball assignment
4. **Physics-based Validation:** Apply golf impact physics for consistency checking

Acceptance Criteria:

- $\geq 80\%$ correct classification on wedge shots (most challenging)
- $\geq 90\%$ correct classification on iron and driver shots
- Velocity separation error < 10 mph RMS vs TrackMan reference
- Impact timing detection within ± 2 ms of theoretical timing
- Handle velocity range 56.5-178.9 mph consistently across all club types

MATHEMATICAL FRAMEWORK

Impact Detection:

$Energy(t) = \sum |STFT(f,t)|^2$ across frequency bins

$\nabla Energy(t) = [Energy(t+\Delta t) - Energy(t-\Delta t)] / 2\Delta t$

$Impact_detected \equiv \nabla Energy(t) > \alpha \times \sigma_{noise}$ ($\alpha \approx 7.0$)

Multi-channel fusion: $E_fused(t) = \sum w_c \times E_c(t)$, $w_c = [0.25, 0.25, 0.25, 0.25]$

Velocity Signature Classification:

$Continuity_metric = (1/N) \times \sum |v(t+1) - v(t)|$

Club: $C < 2.0$ mph/ms (smooth motion)

Ball: $C > 15.0$ mph/ms (impulsive motion)

$Peak_velocity_ratio = \max(v_ball) / \max(v_club)$

Expected ratios: Wedge(1.2-1.8), Iron(1.4-2.2), Driver(1.6-2.5)

Spectral Analysis:

$Spectral_width = \sqrt{[\sum f^2 |X(f)|^2] / \sum |X(f)|^2}$

Club: Broader spectrum (extended contact) > 50 Hz

Ball: Narrower spectrum (point target) < 20 Hz

Cross-correlation:

$R_{cb}(\tau) = \sum v_club(t) \times v_ball(t+\tau) / \sqrt{[\sum v_club^2 \times \sum v_ball^2]}$

Expected delay: 1-3 ms for golf impact physics

Feature Vector (7-dimensional):

Feature $\equiv [max_velocity, rise_time, spectral_width, impact_timing, continuity_metric, duration, correlation_delay]$

REQUIRED FOLDER STRUCTURE

```
milestone3/
├── main_milestone3.m ..... % Main execution script
├── config/
│   ├── milestone3_config.json ..... % All algorithm parameters
│   ├── classification_params.json ..... % ML model configuration
│   └── club_type_configs.json ..... % Wedge/iron/driver specific settings
```

```

└── src/
├── data_io/
│   ├── load_iq_data.m          % Binary I/Q file reader (self-contained)
│   ├── load_json_metadata.m    % JSON parser with TrackMan extraction
│   ├── extract_trackman_reference.m % Reference speed extraction from filenames
│   └── save_separation_results.m % Output management and validation
├── signal_processing/
│   ├── compute_stft_spectrogram.m % STFT with optimized parameters
│   ├── basic_peak_detection.m     % Simple peak detection for tracks
│   └── doppler_to_velocity.m      % Frequency to velocity conversion
├── impact_detection/
│   ├── detect_impact_events.m    % Energy burst detection algorithm
│   ├── temporal_gradient_analysis.m % Energy gradient computation
│   ├── multi_channel_fusion.m    % 4-channel energy fusion
│   └── validate_impact_timing.m   % Physics-based timing validation
├── feature_extraction/
│   ├── extract_velocity_features.m % Peak velocity, rise time, continuity
│   ├── extract_spectral_features.m % Spectral width, bandwidth analysis
│   ├── extract_temporal_features.m % Duration, timing relative to impact
│   ├── compute_cross_correlation.m % Club-ball velocity correlation
│   └── build_feature_vectors.m    % Combine all features into vectors
├── classification/
│   ├── rule_based_classifier.m    % Physics-based decision tree
│   ├── train_ml_classifier.m      % SVM/Random Forest training
│   ├── apply_classification.m     % Execute club/ball classification
│   ├── classification_confidence.m % Uncertainty quantification
│   └── ensemble_classifier.m      % Combine rule-based + ML methods
├── separation_logic/
│   ├── temporal_separation.m      % Time-domain club/ball separation
│   ├── velocity_assignment.m      % Assign velocities to club vs ball
│   ├── physics_validation.m       % Golf ball impact physics checks
│   └── consistency_checks.m       % Cross-validation between methods
├── validation/
│   ├── compute_separation_metrics.m % Classification accuracy, RMS errors
│   ├── trackman_validation.m       % Direct TrackMan comparison
│   ├── confusion_matrix_analysis.m % Error pattern analysis
│   └── performance_by_club_type.m  % Wedge/iron/driver breakdown
├── visualization/
│   ├── plot_impact_detection.m    % Energy burst and timing plots
│   ├── plot_feature_analysis.m    % Feature space visualization, PCA
│   ├── plot_classification_results.m % Confusion matrices, ROC curves
│   ├── plot_velocity_separation.m % Club/ball separation results
│   └── plot_performance_dashboard.m % Complete system performance
└── results/

```

```

| | — impact_detection_results.mat ... % Impact timing and energy data
| | — feature_extraction_results.mat ... % 7D feature vectors for all tracks
| | — classification_results.mat      % Club/ball assignments with confidence
| | — separation_validation.mat       % Velocity separation vs TrackMan
| | — performance_metrics.mat        % Complete accuracy and error statistics
| — models/
| | — trained_svm_classifier.mat     % Trained SVM model
| | — random_forest_model.mat       % Trained Random Forest model
| | — optimized_thresholds.mat      % Rule-based classification thresholds
| — figures/ ... % All visualization outputs organized by type

```

IMPLEMENTATION REQUIREMENTS

1. Self-Contained Data Processing (src/data_io/) Create complete I/Q processing pipeline independent of previous milestones:

- `load_iq_data.m`: Read 4-channel binary I/Q data with proper scaling
- `load_json_metadata.m`: Parse JSON configuration files
- `extract_trackman_reference.m`: Extract ball/club speeds from directory names (LOG format)
- Handle all 130+ dataset files with robust error handling

2. Signal Processing Foundation (src/signal_processing/) Build essential signal processing for club/ball separation:

- `compute_stft_spectrogram.m`: Sliding window STFT optimized for impact detection
- `basic_peak_detection.m`: Simple peak finding in velocity domain
- `doppler_to_velocity.m`: Convert Doppler frequencies to mph using speed coefficient
- Focus on temporal resolution needed for impact event analysis

3. Impact Detection System (src/impact_detection/) Implement robust impact moment identification:

- `detect_impact_events.m`: Energy burst detection using temporal gradients
- `temporal_gradient_analysis.m`: Compute $\nabla \text{Energy}(t)$ with adaptive thresholds
- `multi_channel_fusion.m`: Combine 4-channel I/Q for robust detection
- `validate_impact_timing.m`: Physics-based validation (0.3-3ms impact duration)
- Target: $\pm 2\text{ms}$ accuracy for impact timing detection

4. Feature Extraction Framework (src/feature_extraction/) Extract discriminative features for club vs ball classification:

- `extract_velocity_features.m`: Max velocity, rise time, continuity metrics
- `extract_spectral_features.m`: Spectral width, bandwidth characteristics
- `extract_temporal_features.m`: Duration, timing relative to impact event
- `compute_cross_correlation.m`: Club-ball velocity relationship analysis
- `build_feature_vectors.m`: Construct 7-dimensional feature vectors

5. Classification System (`src/classification/`) Implement hybrid classification approach:

- `rule_based_classifier.m`: Physics-based decision tree (fast, interpretable)
- `train_ml_classifier.m`: SVM (RBF kernel, $\gamma=0.1$, $C=1.0$) and Random Forest training
- `apply_classification.m`: Execute classification with confidence scoring
- `ensemble_classifier.m`: Combine rule-based + ML for robustness
- Target: 80% wedge accuracy, 90% iron/driver accuracy

6. Separation Logic (`src/separation_logic/`) Apply classification to separate club/ball velocities:

- `temporal_separation.m`: Time-domain analysis around impact events
- `velocity_assignment.m`: Assign velocity measurements to club or ball tracks
- `physics_validation.m`: Validate against golf ball impact physics
- `consistency_checks.m`: Cross-validate separation results
- Target: <10 mph RMS error vs TrackMan

7. Validation and Performance Analysis (`src/validation/`) Comprehensive performance evaluation:

- `compute_separation_metrics.m`: Classification accuracy, precision, recall, F1
- `trackman_validation.m`: Direct comparison with reference measurements
- `confusion_matrix_analysis.m`: Error pattern analysis by club type
- `performance_by_club_type.m`: Detailed wedge/iron/driver breakdown

8. Visualization Suite (`src/visualization/`) Create comprehensive analysis plots:

- `plot_impact_detection.m`: Energy burst detection and timing validation
- `plot_feature_analysis.m`: Feature distributions, PCA, correlation matrices
- `plot_classification_results.m`: Confusion matrices, ROC curves, precision-recall
- `plot_velocity_separation.m`: Club/ball separation accuracy vs TrackMan
- `plot_performance_dashboard.m`: Complete system performance summary

CONFIGURATION MANAGEMENT

config/milestone3_config.json:

```
json
```

```
{
  "signal_processing": {
    "sampling_rate": 22700,
    "fft_size": 1024,
    "overlap_percent": 75,
    "window_type": "hamming",
    "zero_padding_factor": 2
  },
  "impact_detection": {
    "energy_gradient_threshold": 7.0,
    "analysis_window_ms": 20,
    "min_impact_duration_ms": 0.3,
    "channel_fusion_weights": [0.25, 0.25, 0.25, 0.25]
  },
  "feature_extraction": {
    "velocity_window_ms": 100,
    "continuity_smooth_threshold": 2.0,
    "continuity_impulsive_threshold": 15.0,
    "rise_time_threshold_ms": 2.0,
    "spectral_width_broad_hz": 50,
    "spectral_width_narrow_hz": 20,
    "correlation_lag_range_ms": 20
  },
  "classification": {
    "rule_based": {
      "velocity_ratio_wedge": [1.2, 1.8],
      "velocity_ratio_iron": [1.4, 2.2],
      "velocity_ratio_driver": [1.6, 2.5]
    },
    "svm": {
      "kernel": "rbf",
      "gamma": 0.1,
      "C": 1.0
    },
    "random_forest": {
      "n_estimators": 100,
      "max_depth": 10
    },
    "confidence_threshold": 0.7
  },
  "validation": {
    "target_wedge_accuracy": 0.8,
    "target_iron_accuracy": 0.9
  }
}
```

```
..... "target_driver_accuracy": 0.9,  
..... "max_velocity_error_mph": 10.0,  
..... "impact_timing_tolerance_ms": 2.0  
..... }  
..... }
```

KEY PERFORMANCE TARGETS

Classification Performance:

- Wedge shots: $\geq 80\%$ correct club/ball classification (primary challenge)
- Iron shots: $\geq 90\%$ correct club/ball classification
- Driver shots: $\geq 90\%$ correct club/ball classification
- Overall dataset: $\geq 85\%$ correct classification with confidence scoring

Velocity Separation Accuracy:

- Club velocity error: < 10 mph RMS vs TrackMan
- Ball velocity error: < 10 mph RMS vs TrackMan
- Impact timing accuracy: ± 2 ms of theoretical impact moment
- Consistent performance across 56.5-178.9 mph velocity range

Feature Discrimination:

- Clear separation in 7D feature space between club and ball tracks
- Robust feature extraction across all shot types and noise conditions
- Feature importance analysis for algorithm optimization

MAIN EXECUTION FLOW

main_milestone3.m structure:

1. Load configuration from JSON files and set up processing parameters
2. Initialize data structures and processing variables
3. Process all dataset files (130+ shots) in batch mode:
 - Load I/Q data and extract TrackMan reference speeds
 - Compute STFT spectrogram optimized for impact detection
 - Detect impact events using multi-channel energy fusion
 - Extract 7D feature vectors from velocity and spectral characteristics

- Apply hybrid classification (rule-based + ML) for club/ball assignment
 - Separate club and ball velocities using temporal analysis
 - Validate results against TrackMan reference measurements
4. Compute comprehensive performance metrics across all shots
 5. Generate complete visualization suite (impact detection, classification, separation)
 6. Save all results, trained models, and performance statistics
 7. Generate automated performance report with recommendations

CRITICAL IMPLEMENTATION NOTES

Physics-Based Constraints:

- Golf ball impact duration: 0.3-3 ms (validate detected impact timing)
- Velocity ratios: Use club-type specific ranges for classification
- Energy conservation: Ball kinetic energy must be consistent with club energy transfer
- Temporal causality: Club peak must occur before or simultaneous with ball peak

Algorithm Robustness:

- Handle edge cases: mishits, partial contact, practice swings, multiple objects
- Implement parameter bounds checking and validation
- Add comprehensive error handling with informative messages
- Memory-efficient processing for large dataset (130+ files)

Classification Strategy:

- Primary: Rule-based classification using physics constraints (fast, interpretable)
- Secondary: Machine learning for complex/ambiguous cases (SVM + Random Forest)
- Ensemble: Combine both approaches with confidence weighting
- Validation: Cross-validation and TrackMan reference comparison

Embedded Deployment Readiness:

- Efficient algorithms suitable for real-time processing
- Fixed-point arithmetic compatibility where possible
- Memory usage optimization for MCU constraints (Renesas/STM32U)
- Computational complexity analysis and optimization

OUTPUT REQUIREMENTS

Quantitative Results:

- Classification confusion matrices by club type (wedge/iron/driver)
- Velocity separation RMS errors vs TrackMan reference
- Impact detection timing accuracy and reliability statistics
- Feature importance rankings and discrimination analysis
- Algorithm computational performance and timing metrics

Visualizations:

- Impact detection: Energy burst plots with detected timing
- Feature analysis: 7D feature space visualization, PCA, distributions
- Classification: Confusion matrices, ROC curves, precision-recall analysis
- Separation results: Club/ball velocity scatter plots vs TrackMan
- Performance dashboard: Complete system accuracy and error analysis

Models and Parameters:

- Trained SVM and Random Forest classification models
- Optimized rule-based classification thresholds
- Feature extraction parameters and sensitivity analysis
- Algorithm configuration recommendations for different scenarios

Documentation:

- Complete algorithm specification with mathematical derivations
- Parameter sensitivity analysis and optimization guidelines
- Performance validation methodology and statistical significance
- Integration recommendations for Milestone 4 (wedge-specific optimization)

SUCCESS VALIDATION

The implementation is successful when:

1. $\geq 80\%$ wedge classification accuracy demonstrated on challenging low-velocity shots
2. $\geq 90\%$ iron/driver classification accuracy with robust performance
3. < 10 mph velocity separation RMS error vs TrackMan across all club types

4. ± 2 ms impact timing detection accuracy with physics validation
5. 7D feature vectors show clear club/ball discrimination in feature space
6. Hybrid classification system (rule-based + ML) operates reliably
7. Complete visualization suite demonstrates all performance aspects
8. Self-contained codebase with comprehensive error handling and validation

Focus on creating production-quality, physics-informed MATLAB code that solves the core challenge of club/ball separation through robust impact detection, feature-based classification, and temporal analysis. The implementation must demonstrate significant advancement toward the ultimate ± 1 mph accuracy target while maintaining computational efficiency for embedded deployment.